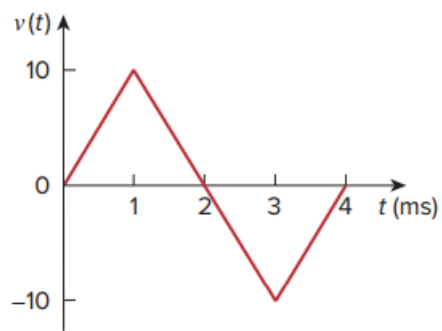


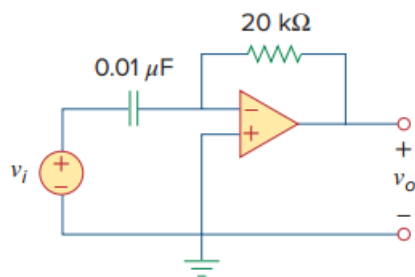
Problem

The triangular waveform in Fig. 6.91(a) is applied to the input of the op amp differentiator in Fig. 6.91(b). Plot the output.

Fig. 6.91:



(a)



(b)

Step-by-step solution

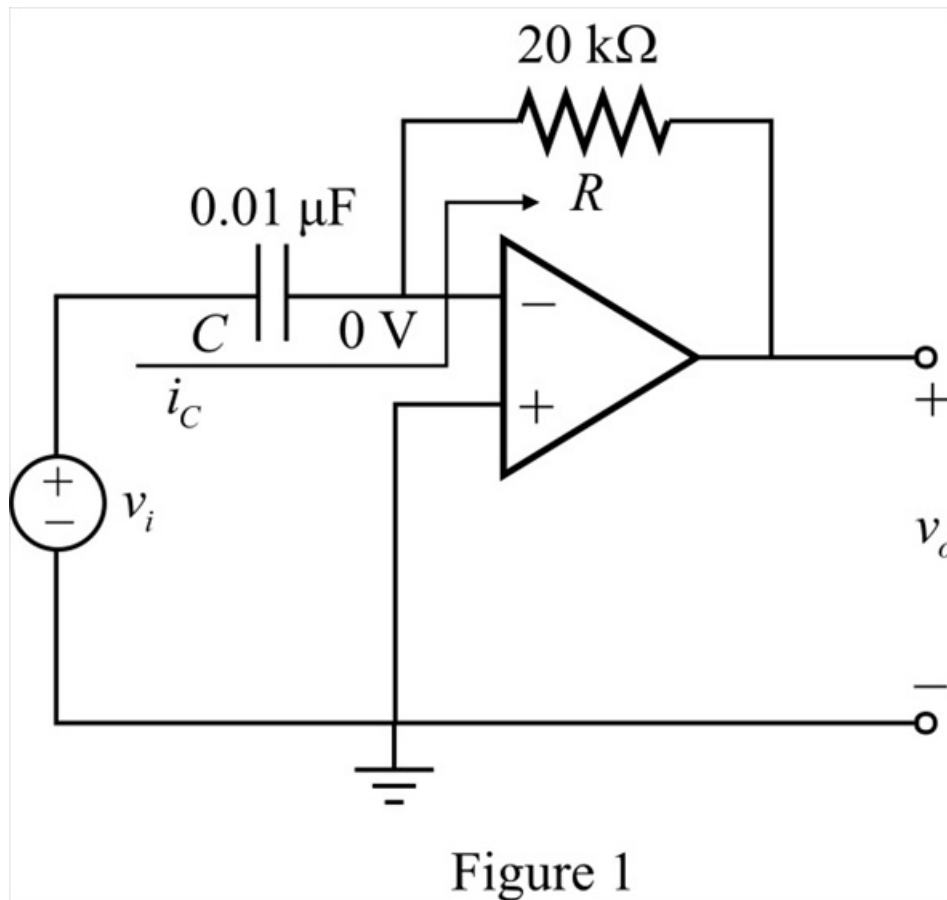
Step 1 of 4

Refer to Figure 6.91 in the textbook.

Refer to Figure 6.91(a) in the textbook for input voltage source.

$$v_i = \begin{cases} 10t & 0 \leq t \leq 1 \\ 20 - 10t & 1 \leq t \leq 3 \\ 10t - 40 & 3 \leq t \leq 4 \\ 0 & \text{otherwise} \end{cases}$$

Draw the modified circuit diagram.



Step 2 of 4

Apply nodal analysis at the inverting terminal.

$$i_C = \frac{0 - v_o}{R}$$

$$C \frac{d(v_i - 0)}{dt} = -\frac{v_o}{R}$$

$$C \frac{dv_i}{dt} = -\frac{v_o}{R}$$

$$v_o = -RC \frac{dv_i}{dt}$$

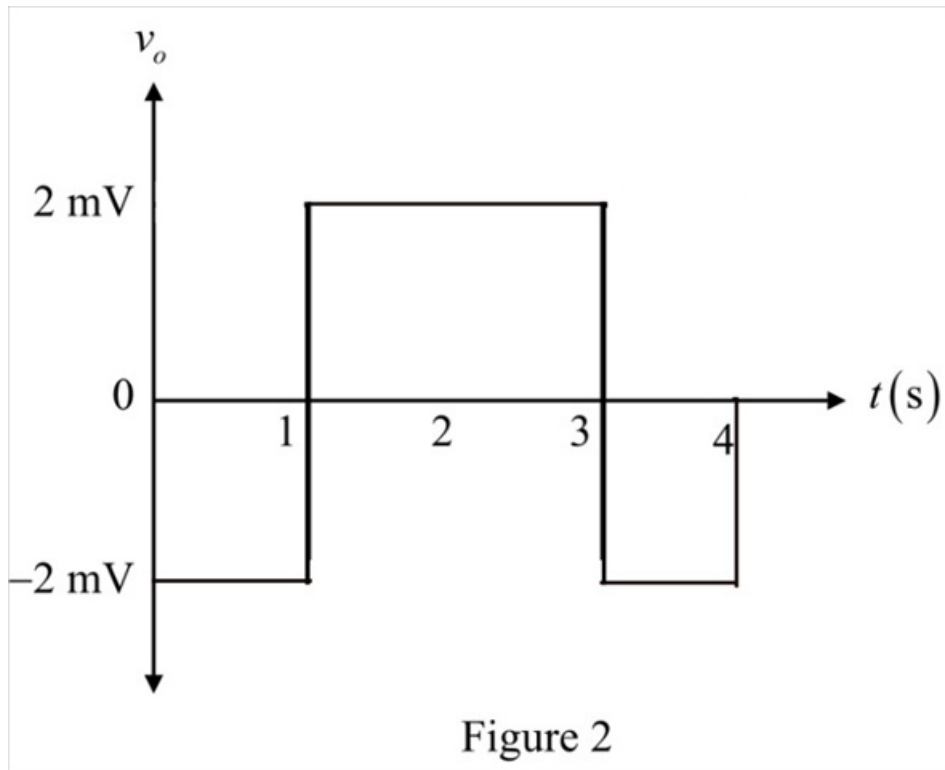
Step 3 of 4

Substitute $20 \text{ k}\Omega$ for R and $0.01 \mu\text{F}$ for C .

$$\begin{aligned}
 v_o &= -(20 \times 10^3)(0.01 \times 10^{-6}) \frac{dv_i}{dt} \\
 &= -(2 \times 10^{-4}) \frac{dv_i}{dt} \\
 &= -(2 \times 10^{-4}) \begin{cases} \frac{d(10t)}{dt} & 0 \leq t \leq 1 \\ \frac{d(20-10t)}{dt} & 1 \leq t \leq 3 \\ \frac{d(10t-40)}{dt} & 3 \leq t \leq 4 \\ 0 & \text{otherwise} \end{cases} \\
 &= -(2 \times 10^{-4}) \begin{cases} 10 & 0 \leq t \leq 1 \\ -10 & 1 \leq t \leq 3 \\ 10 & 3 \leq t \leq 4 \\ 0 & \text{otherwise} \end{cases} \\
 &= \begin{cases} -2 \text{ mV} & 0 \leq t \leq 1 \\ 2 \text{ mV} & 1 \leq t \leq 3 \\ -2 \text{ mV} & 3 \leq t \leq 4 \\ 0 & \text{otherwise} \end{cases}
 \end{aligned}$$

Step 4 of 4

Draw the waveform for output voltage.



Thus, the waveform for output voltage is shown in Figure 2.